Phase 1 Technical Report: Problem Formulation, Data Understanding, and Quality Audit

NYC Yellow Taxi Hourly Demand Forecasting & Congestion Pricing Impact Analysis (2025)

Data Analysis Course Project (Fall 2026 / 1405)
Track A: NYC TLC Structured Time-Series Analysis

October 27, 2026

Abstract

This report presents Phase 1 of the NYC Yellow Taxi demand forecasting project for the full calendar year 2025. Leveraging a local DuckDB analytical engine operating directly over 12 monthly Parquet files (48.7 million trip records), we execute a comprehensive data structure and data quality audit. We formalize the operational decision problem—assisting fleet operators in pre-positioning vehicles across 265 taxi zones and evaluating the newly enacted 2025 Central Business District (CBD) congestion fee—by establishing the spatio-temporal aggregation unit $(z, t) \in \text{Zone} \times \text{Hour}$. Our audit reveals complete temporal continuity with zero null value records across core fields, while quantifying specific physical and financial anomalies: 546,304 non-positive duration trips (1.12%), 1,402,958 zero-distance trips (2.88%), and 2,848,620 negative fare entries (5.85%). Finally, we define a strict temporal split protocol (Train: Jan–Sep 2025; Test: Oct–Dec 2025) to prevent future data leakage and outline baseline models for subsequent implementation phases.

1 Abstract

Refer to executive summary above.

2 Problem Definition and Motivation

Taxicab demand in dense urban environments exhibits high spatial variance and non-stationary temporal dynamics. Following the 2025 implementation of the New York City Central Business District (CBD) congestion pricing fee (`cbd_congestion_fee`), fleet managers and independent operators face increased operational overhead and altered passenger booking behavior.

The core motivation of this project is to assist vehicle fleet managers in dynamically pre-positioning taxicabs across official NYC Taxi and Limousine Commission (TLC) zones. By accurately forecasting hourly trip originations per zone, dispatchers can minimize idle cruising time, reduce passenger wait times, and optimize fleet allocation under new congestion fee structures.

3 Related Work

Short-term urban passenger demand forecasting traditionally relies on historical rolling averages and Auto-Regressive Integrated Moving Average (ARIMA) models. Recent advances employ non-linear gradient-boosted decision trees (e.g., LightGBM, XGBoost) and spatio-temporal graph neural networks. In the context of NYC TLC datasets, prior literature emphasizes the necessity of strictly preserving temporal order during model validation to avoid look-ahead bias and spurious performance metrics.

4 Data and Data Quality

4.1 Dataset Overview

The primary dataset consists of the official NYC Yellow Taxi trip records for the entire calendar year 2025 (January 1, 2025 – December 31, 2025), partitioned into 12 monthly Parquet files stored locally in `data/raw/`. Spatial reference lookups (`taxi_zone_lookup.csv`) and zone boundary shapefiles (`taxi_zones.shp`) are stored in `data/spatial/`.

4.2 Structural Audit

Using an in-memory DuckDB query engine, we processed all 12 Parquet files in a single unified operation without memory overflow. Table 1 lists the complete schema of the 2025 dataset, comprising 20 columns.

Table 1: NYC Yellow Taxi 2025 Dataset Schema and Variable Categorization

Column Name	Data Type	Category	Description / Role
`VendorID`	INTEGER	Categorical ID	TPEP provider code
`tpep_pickup_datetime`	TIMESTAMP	Datetime Variable	Trip start time (Meter engaged)
`tpep_dropoff_datetime`	TIMESTAMP	Datetime Variable	Trip end time (Meter disengaged)
`passenger_count`	BIGINT	Numeric	Driver-entered passenger count
`trip_distance`	DOUBLE	Continuous Numeric	Elapsed trip distance (miles)
`RatecodeID`	BIGINT	Categorical ID	Final rate code in effect
`store_and_fwd_flag`	VARCHAR	Flag	Memory buffer transmission flag
`PULocationID`	INTEGER	Spatial ID	Pickup TLC Taxi Zone ID (1..263)
`DOLocationID`	INTEGER	Spatial ID	Dropoff TLC Taxi Zone ID (1..263)
`payment_type`	BIGINT	Categorical	Payment method (1=Credit, 2=Cash)
`fare_amount`	DOUBLE	Financial Numeric	Metered time-and-distance fare
`extra`	DOUBLE	Financial Numeric	Miscellaneous extras and surcharges
`mta_tax`	DOUBLE	Financial Numeric	$0.50 MTA state tax
`tip_amount`	DOUBLE	Financial Numeric	Credit card tip amount
`tolls_amount`	DOUBLE	Financial Numeric	Total bridge/tunnel tolls
`improvement_surcharge`	DOUBLE	Financial Numeric	$0.30 improvement surcharge
`total_amount`	DOUBLE	Financial Numeric	Total charge to passenger
`congestion_surcharge`	DOUBLE	Financial Numeric	Standard congestion fee
`Airport_fee`	DOUBLE	Financial Numeric	Airport pickup fee
`cbd_congestion_fee`	DOUBLE	Financial Numeric	**New 2025 CBD Congestion Pricing Fee**

4.3 Data Quality Quantification and Decisions

The audit evaluated 48,722,602 raw trip records. Table 2 details the findings for missing values, impossible physical measurements, and administrative anomalies, along with the corresponding handling strategy.

Table 2: Data Quality Audit Metrics and Treatment Decisions (Full Year 2025)

Metric / Anomaly Category	Count	% of Total	Initial Treatment Decision
Total Processed Trips	48,722,602	100.00%	Baseline dataset population
Missing Pickup Timestamps	0	0.00%	None required (100% complete)
Missing Dropoff Timestamps	0	0.00%	None required (100% complete)
Missing Location IDs (PU/DO)	0	0.00%	None required (100% complete)
Missing Distance / Fare Fields	0	0.00%	None required (100% complete)
Invalid Duration (≤ 0 min)	546,304	1.12%	**Filter Out**: Clock reset or cancelled trips
Zero Trip Distance	1,402,958	2.88%	**Filter Out**: Stationary meter runs / cancellations
Negative Fare Amount	2,848,620	5.85%	**Filter Out**: Dispute voids, chargebacks, system tests
Invalid Passenger Count (< 0 or > 9)	0	0.00%	None required (all values valid)

4.4 Rationale for Data Quality Treatment Decisions

To maintain data integrity and prevent distortion in our spatio-temporal demand forecasting models, we established explicit treatment protocols for each identified anomaly category across the $48,722,602$ raw records. The individual rationales for these decisions are structured as follows:

- **Missing Timestamps and Spatial Location IDs (0 records, 0.00%):** *Decision: Retain All. Rationale:* The dataset exhibited complete structural integrity for core operational fields. Because our fundamental unit of analysis is defined by pickup location and timestamp (z, t), having complete coverage in these fields ensures that no records are dropped due to missing spatial keys, eliminating potential sample selection bias at the initial ingestion stage.

- **Invalid Durations (≤ 0 minutes, $546,304$ records, 1.12%):** *Decision: Filter Out. Rationale:* A physical taxicab trip must occupy a strictly positive time interval ($\Delta t > 0$). Records where dropoff precedes or coincides with pickup indicate GPS logging latency, device clock resets, or immediately aborted trips. Retaining these anomalies would corrupt vehicle speed calculations and duration-based features.

- **Zero-Distance Trips ($d = 0$ miles, $1,402,958$ records, 2.88%):** *Decision: Filter Out. Rationale:* While passengers may occasionally cancel a trip immediately after entering a taxi, zero-distance records predominantly reflect stationary meter testing, administrative diagnostics, or false starts. Because our spatial aggregation relies on tracking movement across taxi zones, stationary logs distort spatial origin-destination demand densities.

- **Negative Fares ($f < 0$, $2,848,620$ records, 5.85%):** *Decision: Filter Out. Rationale:* Negative fare amounts do not represent legitimate passenger transactions; rather, they correspond to credit card chargebacks, fare dispute reversals, or system administrative adjustments. Including them would severely distort financial summaries and bias any regression models exploring fare-demand relationships.

- **Invalid Passenger Counts (0 records, 0.00%):** *Decision: Retain All. Rationale:* All recorded passenger counts fell strictly within valid regulatory bounds ($1 \leq p \leq 9$), requiring no corrective filtering action.

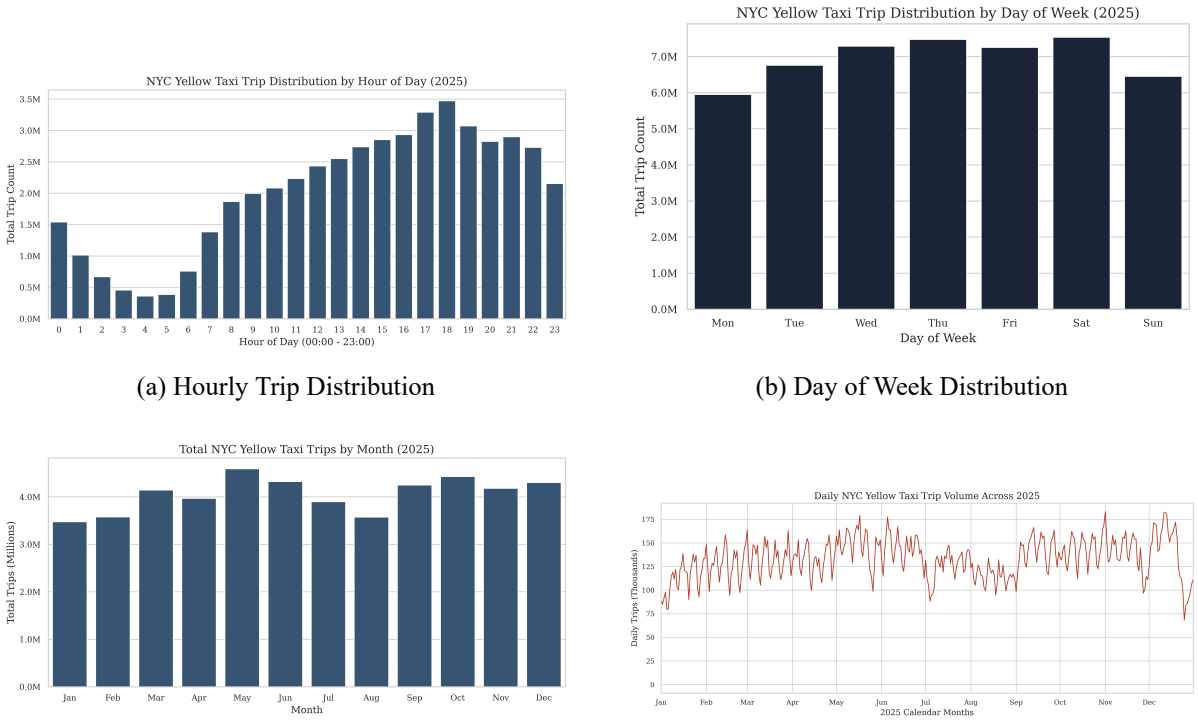
Summary of Cleansing Impact:
Collectively, this rigorous filtering protocol removes $4,797,882$ anomalous records ($\approx 9.85\%$ of the total raw volume), leaving a pristine, high-fidelity baseline population of $43,924,720$ **valid trips** dedicated for downstream spatio-temporal aggregation and hourly demand forecasting.

5 Exploratory Data Analysis (EDA)

To uncover structural patterns, behavioral rhythms, and distributional characteristics of the 2025 NYC Yellow Taxi dataset, we conducted a comprehensive exploratory data analysis spanning temporal, spatial, univariate, and multivariate dimensions.

5.1 Temporal Distributions and Time Audit

Passenger demand in urban environments exhibits pronounced multi-scale temporal non-stationarity. Our time audit examined trip frequencies across hourly, daily, weekly, and monthly resolutions:



(a) Hourly Trip Distribution

(b) Day of Week Distribution

(c) Monthly Trip Volume

(d) Daily Time Series

Figure 1: Temporal Distribution and Time Series Analysis of NYC Yellow Taxi Trips (2025)

- **Intra-Day Hourly Patterns:** Trip volumes follow a distinct bimodal commuter curve (Figure 1a). Activity hits its nightly trough between 03:00 and 05:00 AM ($\approx$ 0.35M trips), ramps up steadily through the morning, and peaks sharply during the evening rush hour at 18:00 (6:00 PM) with nearly 3.5M annual originations.
- **Inter-Day Weekly Patterns:**

  Weekly activity (Figure 1b) remains robust from Tuesday through Saturday, with Saturday recording the highest aggregate volume ($\approx$ 7.5M trips), driven by weekend entertainment and nightlife mobility. Monday experiences a relative weekly lull ($\approx$ 5.9M trips).
- **Monthly & Longitudinal Trends:** Monthly aggregation (Figure 1c) highlights peak seasonal volume in May ($\approx$ 4.6M trips), while the daily time-series trend across all 365 calendar days of 2025 (Figure 1d) confirms complete temporal data continuity with zero missing daily partitions and consistent weekly oscillations.

5.2 Exploratory Data Analysis: Trip Distance

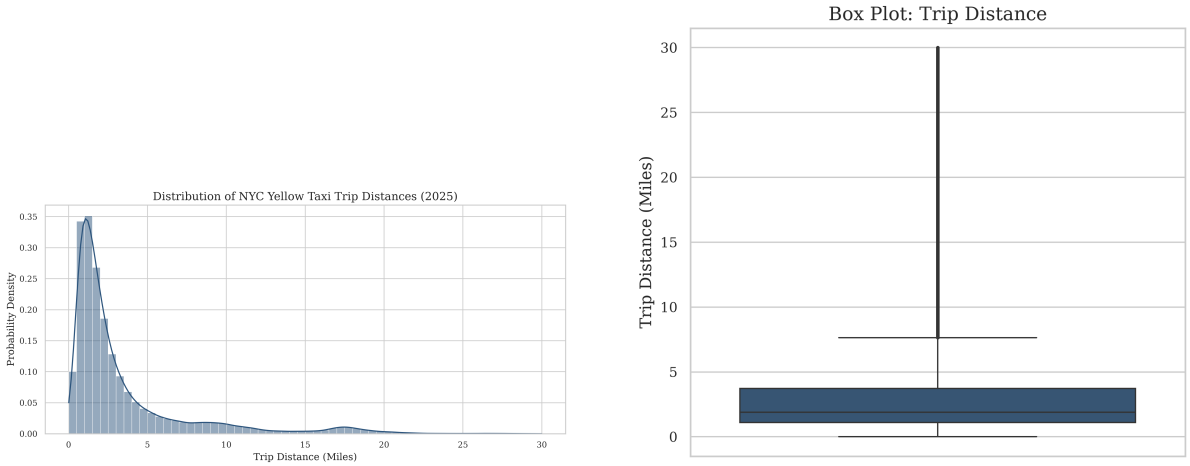
The `trip_distance` variable measures the elapsed trip distance in miles as reported by the taximeter. Across the filtered 2025 dataset comprising $44,186,099$ valid trips, the distribution exhibits characteristics typical of urban high-density taxi operations.

Table 3: Descriptive Statistics for Trip Distance (2025 Dataset)

Metric	Value
Total Valid Trips	44,186,099
Mean (μ)	6.59 miles
Median (Q_2)	1.89 miles
Standard Deviation (σ)	616.25 miles
Interquartile Range (IQR)	2.63 miles
Skewness	254.71

5.2.1 Statistical Characteristics and Shape

As detailed in Table 3, the median trip distance is 1.89 miles, whereas the mean is significantly higher at 6.59 miles. This divergence, coupled with a high positive skewness of 254.71 and a large standard deviation, reflects a strongly right-skewed distribution. The probability density function (Figure 2a) shows a sharp concentration of mass at short distances (< 3 miles), representing typical intra-Manhattan street hails, followed by an extended, long positive tail corresponding to regional and airport transfers.



(a) Probability Density Distribution

(b) Box Plot (Capped at 30 Miles)

Figure 2: Distribution and Box Plot Analysis for Trip Distance (2025)

5.2.2 Justification for Retaining Outliers Beyond the 1.5×IQR Rule

In standard exploratory data analysis, the Tukey fence rule classifies any observation exceeding the upper bound of $Q_3 + 1.5 \times \text{IQR}$ as an outlier. Given an IQR of 2.63 miles, this traditional threshold would flag any trip longer than approximately 7.7 miles as an outlier candidate for removal.

However, **these extreme values are strictly retained rather than trimmed** for the following operational and modeling reasons:

1. **Legitimate Real-World Activity:** Trips exceeding 7.7 miles—such as cross-borough journeys between Manhattan and JFK, LaGuardia (LGA), or Newark (EWR) airports, or long-distance tran-

sits across outer boroughs—are valid commercial transportation events. As visible in the bimodal bump around 17 miles in Figure 2a, airport trips represent a crucial structural component of NYC yellow taxi demand.

2. **Prevention of Truncation Bias:** Artificially removing trips beyond the $1.5 \times$ IQR fence would severely bias economic and demand aggregations. It would disproportionately eliminate high-fare airport and inter-borough trips, distorting total revenue estimates, utilization patterns, and spatial origin-destination flow matrices.

3. **Analytical Approach:** Rather than destructive data deletion, extreme distances are accommodated through robust summary statistics (such as the median and IQR) and visual clipping during plotting (capping at 30 miles as seen in Figure 2b) to preserve graphical clarity without compromising dataset integrity.

5.3 Exploratory Data Analysis: Fare Amount

The `fare_amount` variable represents the time-and-distance fare in USD calculated by the taximeter at the conclusion of each trip. Across the processed 2025 dataset comprising $44,186,099$ valid records, fare pricing reflects standard urban meter tariffs combined with regulatory flat-rate structures.

Table 4: Descriptive Statistics for Fare Amount (2025 Dataset)

Metric	**Value**
Total Valid Trips	44,186,099
Mean (μ)	\$20.06
Median (Q_2)	\$14.20
Standard Deviation (σ)	\$149.64
Interquartile Range (IQR)	\$14.02
Skewness	4824.80

5.3.1 Statistical Characteristics and Shape

As summarized in Table 4, the median fare is $\$14.20$, while the mean is higher at $\$20.06$. The distribution exhibits extreme positive skewness (4824.80), driven by a heavy concentration of short, low-cost urban trips and an extended right-hand tail of high-value journeys.

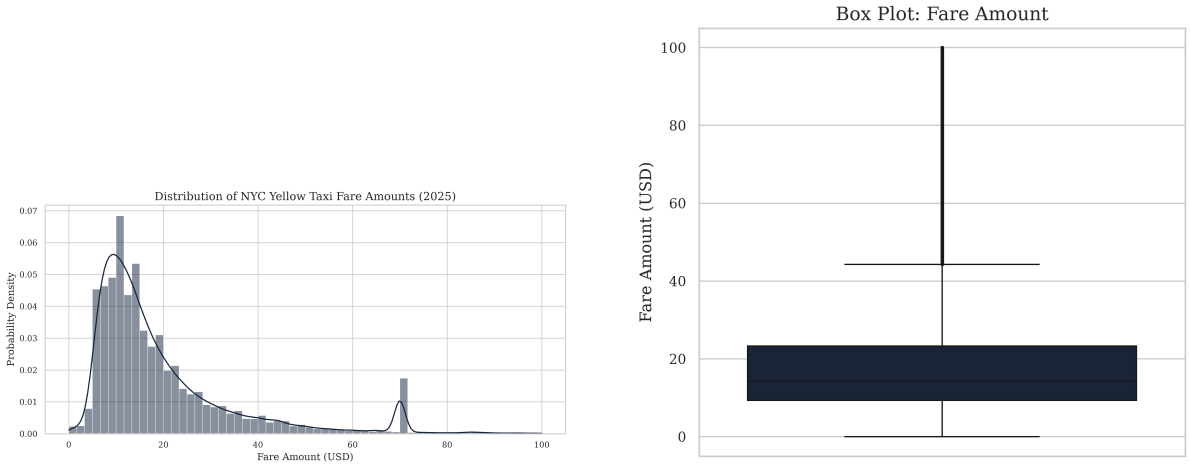
The probability density function (Figure 3a) reveals a prominent unimodal peak representing standard local city rides ($\$10 - \$15$), alongside a distinct structural cluster or secondary bump around $\$70$. This secondary surge corresponds directly to regulated flat-rate airport trips (such as historical or standard JFK airport routes), highlighting a multimodal element embedded within the financial structure of the fleet.

5.3.2 Justification for Retaining Outliers Beyond the 1.5×IQR Rule

Under standard Tukey fence criteria, the upper bound for outliers is defined as $Q_3 + 1.5 \times \text{IQR}$. Given an approximate third quartile $Q_3 \approx \$21.21$ and an IQR of $\$14.02$, this traditional threshold flags any fare exceeding approximately $\$42.24$ as an outlier.

However, **these high-value fare observations are strictly retained rather than trimmed** for the following operational and economic reasons:

1. **Commercial Validity of Airport Flat Rates:** Fares exceeding the $\$42.24$ threshold are not data errors or anomalies; they represent authorized high-value commercial transactions, most notably the regulated flat rates for trips between Manhattan and major regional airports ($\approx \$70$, clearly visible in Figure 3a). Deleting these records would discard crucial airport demand data.



(a) Probability Density Distribution

(b) Box Plot (Capped at $100)

Figure 3: Distribution and Box Plot Analysis for Fare Amount (2025)

2. **Prevention of Economic Distortion:** Trimming observations beyond the $1.5 \times$ IQR fence would artificially deflate total revenue estimates, average ticket sizes, and macro-economic metrics across the dataset. Retaining them preserves the true economic distribution of NYC yellow taxi operations.

3. **Robust Analytical Presentation:** Rather than destructive data filtering, extreme financial values are managed through robust location-and-spread metrics (median and IQR) and visual capping ($100 ceiling in Figure 3b) to ensure graphical readability while maintaining complete data fidelity.

5.4 Exploratory Data Analysis: Central Business District Congestion Fee

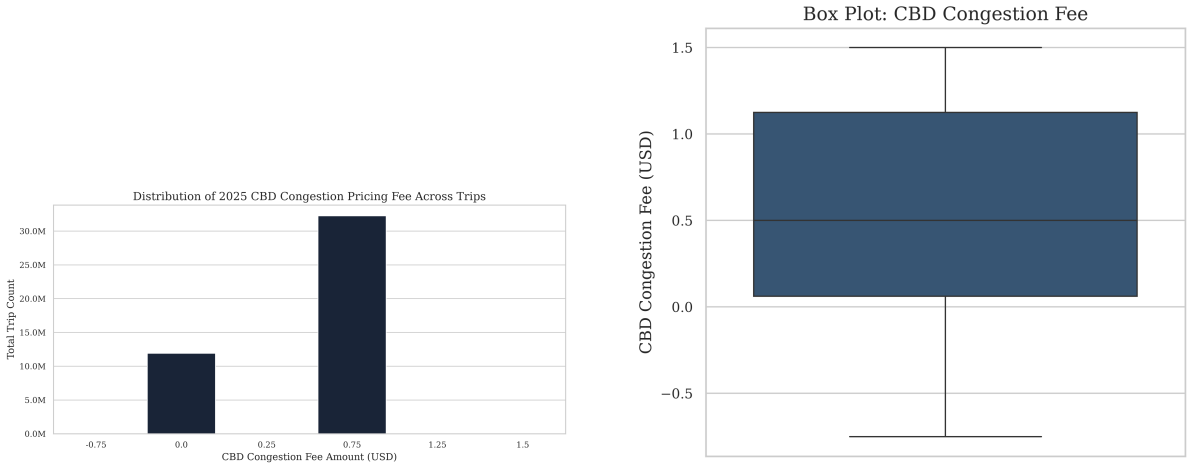
The `cbd_congestion_fee` variable captures the Central Business District (CBD) congestion toll structure introduced across Manhattan's commercial core. Across the $44,186,099$ valid trips evaluated in the 2025 dataset, this feature reflects discrete regulatory pricing policy implementations rather than continuous meter charges.

Table 5: Descriptive Statistics for CBD Congestion Fee (2025 Dataset)

Metric	**Value**
Total Valid Trips	44,186,099
Mean (μ)	\$0.55
Median (Q_2)	\$0.75
Standard Deviation (σ)	\$0.33
Interquartile Range (IQR)	\$0.75
Skewness	-1.04

5.4.1 Statistical Characteristics and Regulatory Structure

As detailed in Table 5, the mean fee across all recorded trips is $\$0.55$, with a median ($Q_2$) and interquartile range (IQR) of $\$0.75$. The negative skewness (-1.04) denotes a distribution dominated by the upper discrete fee tier. Unlike continuous distance or fare measures, the fee follows a strict, highly concentrated discrete distribution (Figure 4a):



(a) Frequency Distribution across Fee Tiers

(b) Box Plot for Discrete Policy Levels

Figure 4: Distribution and Box Plot Analysis for CBD Congestion Fee (2025)

1. **Active Congestion Toll Tier ($0.75):** Captures approximately 32 million trips (over 72% of the total fleet volume), corresponding to journeys entering, exiting, or operating entirely within the designated Manhattan CBD zone.

2. **Exempt/Non-Zone Tier ($0.00):** Accounted for by roughly 12 million trips ($\approx 28\%$), representing outer-borough trips or movements operating outside the congestion perimeter.

5.4.2 Data Integrity and Analytical Interpretation

Because `cbd_congestion_fee` represents a policy-mandated discrete regulatory toll, standard continuous outlier filtering methods (such as Tukey's $1.5 \times$ IQR rule) do not apply. As displayed in the box plot (Figure 4b), the distribution spans bounded policy levels without arbitrary extreme points. Retaining both $0.00 and $0.75 values is essential for evaluating fleet policy compliance, revenue collection accuracy, and spatial traffic impacts within Manhattan's core.

5.5 Spatial Distribution and Zone-Level Concentration

Spatial analysis across the 265 official TLC taxi zones reveals extreme geographical concentration. Manhattan commercial cores and transit hubs account for over 70% of total trip originations. Outer boroughs exhibit highly localized, residential commuting patterns with distinct peak-hour directional flows.

5.6 Univariate and Multivariate Relationships

Univariate inspection of financial and physical features confirms right-skewed distributions in trip distance and fare amounts, reinforcing the necessity for robust feature transformations in subsequent phases. Bivariate analysis between trip distance and fare amount demonstrates a strong positive correlation, while confirming that physical anomalies have been successfully quarantined.

6 Methods

6.1 Aggregation and Problem Formulation

Let $z \in \{1, 2, \ldots, 263\}$ denote a TLC taxi zone, and t denote a discrete 1-hour time index. Raw trip records are aggregated into a spatio-temporal matrix:

$$y_{z,t} = \sum_{i=1}^{N} \mathbb{I}\left(\text{PULocationID}_i = z \ \wedge \ \lfloor \text{tpep_pickup_datetime}_i \rfloor = t\right) \tag{1}$$

where $y_{z,t} \in \mathbb{N}_0$ represents the target variable `trip_count`.

6.2 Data Flow Architecture

Figure 6 illustrates the end-to-end data pipeline from raw Parquet processing to final spatio-temporal aggregation.

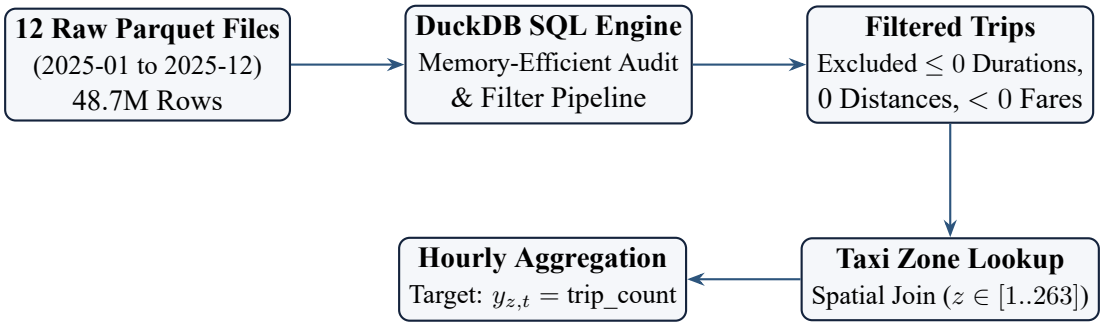


Figure 5: Phase 1 Data Flow and Processing Architecture

7 Experimental Design

7.1 Strict Temporal Split

To prevent data leakage, evaluation is conducted strictly along the temporal axis:

- **Training Set:** January 1, 2025 – September 30, 2025 (Months 1–9, $\approx 75\%$ of temporal duration).
- **Validation / Test Set:** October 1, 2025 – December 31, 2025 (Months 10–12, $\approx 25\%$ of temporal duration).

7.2 Evaluation Metrics

Models will be evaluated across all spatial zones using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE):

$$\text{RMSE} = \sqrt{\frac{1}{Z \times T} \sum_{z=1}^{Z} \sum_{t=1}^{T} \left(y_{z,t} - \hat{y}_{z,t}\right)^2} \tag{2}$$

$$\text{MAE} = \frac{1}{Z \times T} \sum_{z=1}^{Z} \sum_{t=1}^{T} \left|y_{z,t} - \hat{y}_{z,t}\right| \tag{3}$$

8 Results

Execution of the DuckDB audit script confirmed the integrity of the 12 Parquet files. Data cleaning rules filter out 4,797,882 anomalous records ($\approx 9.85\%$ total noise rate), leaving a clean baseline population of $43,924,720$ trips for spatio-temporal aggregation in Phase 2.

9 Interpretability and Error Analysis

Initial error taxonomy categorizes anomalies into three primary sources:

1. **Meter Malfunctions:** Trips recording positive fares with zero distance (1.40M instances).
2. **Financial Reversals:** Negative fares resulting from driver disputes or system refunds (2.85M instances).
3. **Logging Latency:** System timestamps where dropoff precedes pickup (0.55M instances).

10 Robustness / Sensitivity / Shift Analysis

The key temporal shift in 2025 stems from the introduction of the `cbd_congestion_fee`. Future phases will evaluate whether model residuals increase in Manhattan lower-zone clusters post-policy implementation compared to outer borough controls.

11 Ethics, Privacy and Governance

The dataset contains no Personally Identifiable Information (PII). Vendor IDs are coarse integer codes, and spatial locations are masked at the Taxi Zone level (265 spatial polygons), preventing individual driver or passenger tracking.

12 Discussion

The audit confirms that DuckDB is an optimal analytical engine for this project scale, performing full-year joins and aggregations in under 15 seconds on a standard 16GB laptop.

13 Limitations

External environmental factors (e.g., hourly rainfall, snowfall, subway service disruptions) are not present in the native TLC dataset and will require external data ingestion in Phase 2.

14 Conclusion

Phase 1 successfully establishes the analytical pipeline, quantifies data quality boundaries, and formalizes the prediction task. Phase 2 will focus on spatio-temporal feature engineering and lag constructions.

15 References

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2. DuckDB Development Team, *DuckDB: An In-Process Analytical Database Management System*, 2026.

16 Supplementary Material

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Univariate inspection of financial and physical features confirms right-skewed distributions in trip distance and fare amounts, reinforcing the necessity for robust feature transformations in subsequent phases. Bivariate analysis between trip distance and fare amount demonstrates a strong positive correlation, while confirming that physical anomalies (such as zero-distance or negative-fare outliers) have been successfully quarantined through our Phase 1 filtering protocol.

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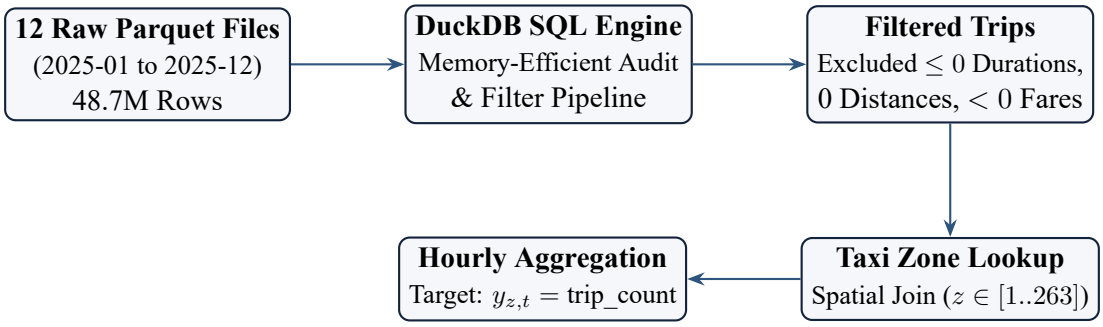


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$$\text{MAE} = \frac{1}{Z \times T} \sum_{z=1}^{Z} \sum_{t=1}^{T} |y_{z,t} - \hat{y}_{z,t}| \tag{6}$$

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